Department of Physics, IIT Madras

PH1020 Physics II **Mid-term Examination** **23.03.2024**
Time: 9:30 am to 11:30 am ***Answer all questions*** **Max. marks: 30 Marks**

Name	Roll No.	Old Roll No.(if any)

Instructions: Each booklet should contain a 5 page question paper attached to an answer booklet containing six pages **(please check this now)**. The answers for **Part A** and **Part B** should be written in the allotted boxes in the question paper, **(for the objective questions we will not evaluate the answers if it is written in any place other than the allotted boxes)**. The answers to **Part C** should be written in the attached answer booklet. Vectors must be indicated clearly using arrows. You may use the reverse side of all pages in the question paper for rough work. All symbols have their usual meaning unless stated otherwise. You must use only black or blue ink for writing the answers and sketching the figures.Calculators, cell phones or any internet connectable device must not be in your possession during the examination.

Exam Hall No.

Total Marks

Useful Formulae

Cylindrical Polar coordinates:

$$\begin{aligned}
\vec{\nabla}u &= \frac{\partial u}{\partial \rho}\,\hat{e}_\rho + \frac{1}{\rho}\frac{\partial u}{\partial \varphi}\,\hat{e}_\varphi + \frac{\partial u}{\partial z}\,\hat{e}_z \\
\vec{\nabla}\cdot\vec{A} &= \frac{1}{\rho}\frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho}\frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z} \\
\vec{\nabla}\times\vec{A} &= \left[\frac{1}{\rho}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right]\hat{e}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho}\right]\hat{e}_\varphi + \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}(\rho A_\varphi) - \frac{\partial A_\rho}{\partial \varphi}\right]\hat{e}_z
\end{aligned}$$

Spherical Polar coordinates:

$$\begin{aligned}
\vec{\nabla}u &= \frac{\partial u}{\partial r}\,\hat{e}_r + \frac{1}{r}\frac{\partial u}{\partial \theta}\,\hat{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial u}{\partial \varphi}\,\hat{e}_\varphi \\
\vec{\nabla}\cdot\vec{A} &= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(A_\theta \sin\theta) + \frac{1}{r\sin\theta}\frac{\partial A_\varphi}{\partial \varphi} \\
\vec{\nabla}\times\vec{A} &= \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial \theta}(\sin\theta A_\varphi) - \frac{\partial A_\theta}{\partial \varphi}\right]\hat{e}_r \\
&\quad + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r}(r A_\varphi)\right]\hat{e}_\theta + \frac{1}{r}\left[\frac{\partial}{\partial r}(r A_\theta) - \frac{\partial A_r}{\partial \theta}\right]\hat{e}_\varphi
\end{aligned}$$

Part A

Clearly indicate your answers to each of the questions below in the boxes provided. The answers to each of these questions **must be a number or an expression.** Each box is worth 1 **mark.** [9 × 1 = 9 marks]

1. The screened Coulomb potential is given by

$$\Phi(r) = \frac{q}{4\pi\epsilon_0}\frac{\exp(-r/\lambda)}{r},$$

where r is the usual radial co-ordinate, and λ is a positive constant.

(a) The charge distribution $\rho(r)$ corresponding to this potential is:

$\rho(r) =$ $-qe^{-r/\lambda}/4\pi r\lambda^2$

(b) The total charge enclosed within a sphere of radius λ, with its origin at $r = 0$ is:

$Q_{enc} =$ $2q/e$

2. A point dipole $\vec{p_1} = p_1\hat{e_z}$ is held fixed at the origin O. Another point dipole $\vec{p_2}$ with magnitude p_2 is located on the z-axis at a distance h above O, free to rotate in all directions. The minimum value of the potential energy for this configuration is:

$U_{min} =$ $-p_1p_2/2\pi\epsilon_0h^3$

3. A square sheet of length L carrying a charge distribution $\sigma(x, y) = kx$, (with k a positive constant) is centred at the origin with its sides parallel to the Cartesian axis. The electric dipole moment of this configuration is:

$\vec{p} =$ $kL^4/12$

4. The electric polarization of a linear, isotropic, and homogenous dielectric material with electric susceptibility χ_e, relative permittivity $\epsilon_r = 3$ and electric displacement $\vec{D} = 4 \times 10^{-7}\hat{e_x}$ C/m^2 is:

$\vec{P} =$

5. A sphere of radius R contains a dielectric material that carries a polarization given by

$$\vec{\mathbf{P}} = \frac{k}{r}\hat{\mathbf{e}}_{\mathbf{r}},$$

with k a positive constant.

(a) The bound volume charge density in the medium is given by:

$\rho_b =$

(b) The bound surface charge density at $r = R$ is given by:

$\sigma_b =$

6. Given a charge distribution of the form

$$\rho(r,\theta,\varphi) = \begin{cases} \beta[1-(r^2/a^2)] & \text{if } 0 < r \le a \\ 0 & \text{if } a < r < \infty. \end{cases}$$

Here, β is a constant of appropriate dimensions

(a) The electric field in the region $0 < r \le a$ is

$\vec{\mathbf{E}} =$

(b) The electric field in the region $a < r < \infty$ is

$\vec{\mathbf{E}} =$

Part B

Each of the questions below are worth 2 marks. Pick the correct choice(s) for each of the questions below, and write your choices in the box provided for each question. Further, all constants have appropriate dimensions. Marks will be given only if no wrong choice/choices is/are picked. [3 × 2 = 6 marks]

1. For a single point charge $+q$ placed at $\vec{r} = (1,0,0)$, which of the following statements are correct:
   (A) The dipole moment is finite.
   (B) The monopole moment is finite.
   (C) The dipole moment is independent of the choice of origin.
   (D) The quadrupole moment is zero.

2. For a conductor of arbitrary shape, pick the statements that hold true:
   (A) The electrostatic potential inside a conductor should be strictly zero.
   (B) The electric field on the surface of a conductor must be normal to the surface.
   (C) The mobile charges distribute themselves on the surface because this is the configuration in which total electrostatic energy is minimum.
   (D) If there exists a cavity of arbitrary shape inside the conductor, the net electric field in the conductor is non-zero.

3. It is known that the trajectory of a charged particle (starting from rest at the origin), moving under the combined effect of a crossed electric ($E\hat{e_z}$) and magnetic field ($B\hat{e_x}$) is described by a circle of radius R whose centre $(0, R\omega t, R)$ travels in the y direction, (with ω being a frequency). The following statements hold for R, and ω:

   (A) $\omega \propto B$.
   (B) $\omega \propto 1/B$.
   (C) $R \propto 1/B^2$.
   (D) $R \propto 1/B$.

Part C

These are questions that are subjective in nature. You need to show a clear work out of your solutions for this section in the answer sheets attached to the question paper. In the answer sheets clearly mark the question number that you are answering. All constants have appropriate dimensions. [15 marks]

1. Consider the following potential given by:

$$\Phi = \begin{cases} \frac{\rho_0}{4\pi\epsilon_0}\left(x^2+y^2+z^2\right) & \text{if } x^2+y^2+z^2 < a^2 \\ \frac{\rho_0}{4\pi\epsilon_0}\left(-a^2+\frac{2a^3}{(x^2+y^2+z^2)^{1/2}}\right) & \text{if } x^2+y^2+z^2 > a^2 \end{cases}$$

   Here, ρ_0 has the dimensions of a volume charge density and is a constant. Furthermore, ϵ_0 is the permittivity of free space. [3 marks]

   (a) Calculate and plot the electric field corresponding to the above potential as a function of the radial co-ordinate r.

   (b) Calculate the surface charge density at $r = a$.

2. An infinite slab of a linear isotropic homogenous dielectric material of relative permittivity ϵ_r, and thickness h is in contact with the surface of a conductor of infinite extent carrying a uniform charge density σ. Assume the top surface of the conductor is lying coincident with the xy plane. Find: [4 marks]

   (a) The electric field both inside and outside the slab.

   (b) The bound surface charge densities on both the top and bottom surface of the dielectric.

3. A wire centred at the origin is placed symmetrically along the z-axis. It carries a total charge Q, distributed via a linear charge density given by:

$$\lambda(z) = \begin{cases} \lambda_0 \left(1 - z^2/a^2\right) & \text{if } -a \leq z \leq a \\ 0 & \text{else.} \end{cases}$$

   Here, the λ_0 is a positive constant of appropriate dimensions. Find the approximate potential upto quadrapolar order for the above charge distribution at the point $(0, 0, z)$, (assume $z \gg 2a$).

   [5 marks]

4. A system consists of a long solid inner cylinder of radius a, and a concentric very thin outer cylinder, of radius b. A uniform volume charge density ρ is distributed in the inner cylinder and outer cylinder has an equal but opposite charge pasted on the outer cylinder. Find the electrostatic energy per unit length in the system of cylinders.

   [3 marks]

1. Consider the following potential given by:

$$\Phi = \begin{cases} \frac{\rho_0}{4\pi\epsilon_0}\left(x^2+y^2+z^2\right) & \text{if } x^2+y^2+z^2 < a^2 \\ \frac{\rho_0}{4\pi\epsilon_0}\left(-a^2 + \frac{2a^3}{(x^2+y^2+z^2)^{1/2}}\right) & \text{if } x^2+y^2+z^2 > a^2 \end{cases}$$

Here, ρ_0 has the dimensions of a volume charge density and is a constant. Furthermore, ϵ_0 is the permittivity of free space. [3 marks]

(a) Calculate and plot the electric field corresponding to the above potential as a function of the radial co-ordinate r.

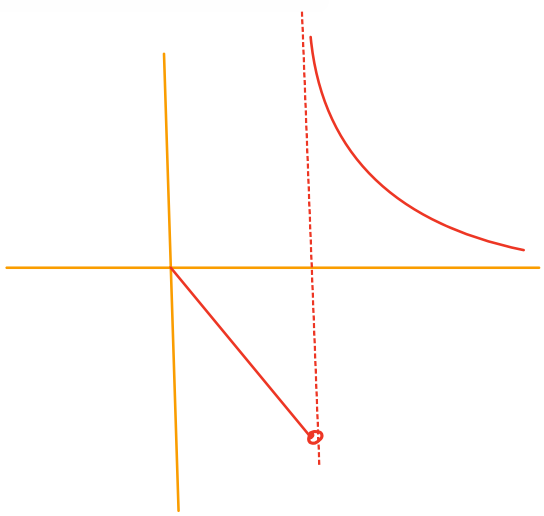
(b) Calculate the surface charge density at $r = a$.

$\phi = \frac{\rho_0}{4\pi\epsilon_0} r^2 \quad r<a$

$= \frac{\rho_0 a^2}{4\pi\epsilon_0}[2a/r - 1] \quad r>a$

$E = -\nabla\phi = -\frac{\rho_0 r}{2\pi\epsilon_0}\hat{r} \quad r<a$

$= \frac{\rho_0 a^3}{2\pi\epsilon_0 r^2}\hat{r} \quad r>a$



$\sigma = \epsilon_0\left[\frac{\rho_0 a}{2\pi\epsilon_0} + \frac{\rho_0 a}{2\pi\epsilon_0}\right] = \frac{\rho_0 a}{\pi}$

$\sigma = -\epsilon\frac{\partial V}{\partial n}$

3. A wire centred at the origin is placed symmetrically along the z-axis. It carries a total charge Q, distributed via a linear charge density given by:

$$\lambda(z) = \begin{cases} \lambda_0\left(1 - z^2/a^2\right) & \text{if } -a \le z \le a \\ 0 & \text{else.} \end{cases}$$

$z^4/a^2 \quad 2z^2/5a^2$

Here, the λ_0 is a positive constant of appropriate dimensions. Find the approximate potential upto quadrapolar order for the above charge distribution at the point $(0, 0, z)$, (assume $z \gg 2a$).

[5 marks]

monopole: $\frac{1}{4\pi\epsilon_0}\frac{\int\lambda dz}{z} = \frac{4a\lambda_0/3}{4\pi\epsilon_0 z} = \frac{a\lambda_0}{3\pi\epsilon_0 z}$

dipole: $\frac{1}{4\pi\epsilon_0}\frac{\int\lambda z dz}{z^2} = 0$

Quadrupole:

$$Q_{zz} = \int \lambda dz\,(3z^2 - z^2)$$

$$= 2\lambda_0\left[\frac{2a^3}{3} - \frac{2a^3}{5}\right] = \frac{8\lambda a^3}{15}$$

$$Q_{xx} = Q_{yy} = \frac{-4\lambda a^3}{15}$$

$$Q = \frac{4\lambda a^3}{15}\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$q = \frac{4a}{3\lambda_0}$$

$$\lambda = \frac{3q}{4a}$$

$$V = \frac{1}{8\pi\varepsilon_0 z^5}\begin{bmatrix} x & y & z \end{bmatrix} Q \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \frac{\lambda a^3}{30\pi\varepsilon_0 z^5}\left[-x^2 - y^2 + 2z^2\right]$$

$$V_Q = \frac{\lambda a^3}{15\pi\varepsilon_0 z^3}$$

$$V = \frac{q\lambda_0}{3\pi\varepsilon_0 z} + \frac{\lambda a^3}{15\pi\varepsilon_0 z^3}$$

$\frac{kQ}{z}$ $\frac{kq}{5}\frac{a^2}{z^3}$

4. A system consists of a long solid inner cylinder of radius a, and a concentric very thin outer cylinder, of radius b. A uniform volume charge density ρ is distributed in the inner cylinder and outer cylinder has an equal but opposite charge pasted on the outer cylinder. Find the electrostatic energy per unit length in the system of cylinders.

[3 marks]

$$E \times 2\pi R\ell = \frac{\rho \times \pi R^2 \ell}{\varepsilon_0} \Rightarrow E = \frac{\rho r}{2\varepsilon_0} \qquad r \in [0, a]$$

$$E \times 2\pi R\ell = \frac{\rho \times \pi a^2 \ell}{\varepsilon_0} \Rightarrow E = \frac{\rho a^2}{2\varepsilon_0 r} \qquad r \in [a, b]$$

$$E = 0 \quad r > b$$

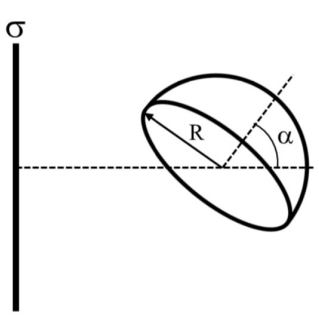
$$U = \tfrac{1}{2}\varepsilon_0 E^2 \Rightarrow U = \int \tfrac{1}{2}\varepsilon_0 E^2 \, 2\pi r \, dr$$

$$= \tfrac{1}{2}\varepsilon_0 \int_0^a \frac{\rho^2 r^2}{4\varepsilon_0^2} \cdot 2\pi r \, dr + \int_a^b \frac{\rho^2 a^4}{4\varepsilon_0^2 r^2} 2\pi r \, dr$$

$$= \frac{1}{2}\varepsilon_0 \left[\frac{\rho^2 \pi}{2\varepsilon_0^2} \frac{a^4}{4} + \frac{\rho^2 a^4 \pi}{2\varepsilon_0^2} \ln(b/a) \right]$$

$$= \frac{\pi \rho^2 a^4}{4\varepsilon_0} \left[1/4 + \ln(b/a) \right]$$

7. An infinite plane conducting sheet has a uniform surface charge density $+\sigma$. A hemispherical Gaussian surface of radius R has its symmetry axis at an angle α to the normal to the infinite plane, as shown in figure. Calculate the flux of electric field passing through the curved surface of the hemisphere.



$$\frac{\sigma}{2\varepsilon_0} \times \pi r^2 \times \cos\alpha$$